

Performance Marketing Analysis — Case Study

Personal Data Analytics Project — Multi-Channel Marketing Campaign Performance Dashboard

Dataset: Public Marketing Campaign Performance Dataset | Tool: Power BI | Role: Data Analyst (self-directed)

1. Project Overview

This is a self-directed practice project built on a public marketing campaign performance dataset, covering campaigns run by multiple companies across several digital channels (Google Ads, Facebook, YouTube, Instagram, and Website) between 2021 campaign cycles. The goal was to build an interactive dashboard that a marketing or growth team could use to evaluate channel performance, campaign efficiency, and audience engagement at a glance.

2. Objective

The dataset contains granular, campaign-level records rather than pre-aggregated summaries, so the core challenge was to clean and structure the raw data, then design a dashboard that answers three practical business questions:

- Which channels and campaign types deliver the best return on investment (ROI)?
- Which customer segments engage most with campaigns, and where are campaigns concentrated geographically?
- Is higher acquisition spend actually associated with higher ROI, or does spend efficiency vary by campaign?

3. Dataset

The raw dataset consists of individual marketing campaign records with the following fields:

Field	Description
Campaign_ID, Company	Unique identifier and the company running the campaign
Campaign_Type, Channel_Used	Format of the campaign (Email, Display, Search, Social Media, Influencer) and the platform used (Google Ads, Facebook, YouTube, Instagram, Website)
Target_Audience, Customer_Segment	Demographic target (e.g. Men 18-24) and behavioral segment (e.g. Tech Enthusiasts, Fashionistas)
Duration, Date	Campaign length in days and start date
Conversion_Rate, Acquisition_Cost, ROI	Core performance metrics per campaign
Clicks, Impressions, Engagement_Score	Traffic and engagement metrics
Location, Language	Market and language targeting for the campaign

A sample of the raw records is shown below:

Company	Type	Channel	Conv. Rate	Cost	ROI	Segment
Innovate Industries	Email	Google Ads	0.04	\$16,174	6.29	Health & Wellness
NexGen Systems	Email	Google Ads	0.12	\$11,566	5.61	Fashionistas
Alpha Innovations	Influencer	YouTube	0.07	\$10,200	7.18	Outdoor Adventurers
DataTech Solutions	Display	YouTube	0.11	\$12,724	5.55	Health & Wellness
Alpha Innovations	Social Media	Facebook	0.09	\$18,066	6.73	Tech Enthusiasts

4. Methodology

Step	What I Did
Data Cleaning	Standardized cost fields (removed currency formatting), verified conversion rate and ROI ranges, and checked for duplicate campaign IDs and missing values.
Metric Design	Derived headline KPIs — Total Clicks, Total Impressions, CTR, CPC, Weighted ROI, and Average Conversion Rate — to summarize performance at a glance.
Dashboard Design	Built interactive filters (Campaign_Type, Channel_Used, Year-Month) so the dashboard can be sliced by format, platform, or time period.
Visualization	Created six visuals covering channel ROI, conversion rate by campaign type, click trend over time, cost-vs-ROI efficiency, customer segment engagement, and geographic distribution of ROI.

5. The Dashboard

The dashboard combines top-level KPIs with six supporting visuals, letting a viewer move from a high-level summary down to channel-, segment-, or location-level detail.

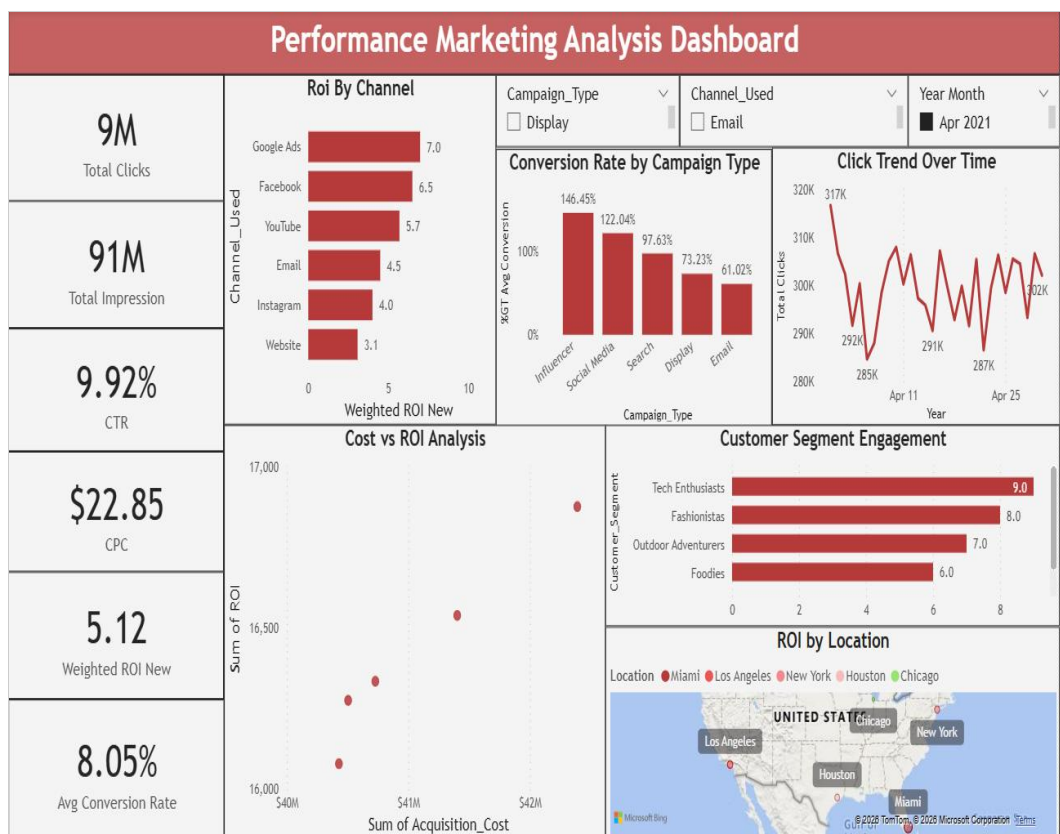


Fig. 1 — Performance Marketing Analysis Dashboard: KPIs, ROI by channel, conversion rate by campaign type, click trend, cost-vs-ROI, customer segment engagement, and ROI by location.

9M TOTAL CLICKS	91M TOTAL IMPRESSIONS	9.92% CTR	\$22.85 CPC	5.12 WEIGHTED ROI	8.05% AVG CONV. RATE
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6. Key Insights

- Google Ads and Facebook lead on ROI:** Google Ads (~7.0) and Facebook (~6.4) return the highest ROI among all channels, ahead of YouTube (~5.7), Instagram (~4.9), and Website (~4.0) — pointing to where paid spend is working hardest.
- Tech Enthusiasts and Fashionistas are the most engaged segments:** these two customer segments post the highest engagement scores, ahead of Outdoor Adventurers and Foodies, suggesting content and offers resonate more strongly with them.
- Cost and ROI are not tightly linked:** the cost-vs-ROI scatter shows campaigns with similar acquisition costs achieving very different ROI outcomes, meaning spend size alone doesn't predict return — campaign quality and targeting matter more.
- Click volume fluctuates month to month:** the click trend line shows no single sustained upward or downward pattern across the sampled period, indicating performance is likely driven more by individual campaign execution than by broad seasonality.
- Campaign activity concentrates in major metros:** Los Angeles, New York, Chicago, and Houston account for the bulk of geographic ROI activity, useful for prioritizing regional budget allocation.

7. Recommendations

- Reallocate incremental budget toward Google Ads and Facebook, the two highest-ROI channels, while testing whether Website campaigns can be improved or phased out.
- Build segment-specific creative and offers for Tech Enthusiasts and Fashionistas, given their consistently higher engagement scores.
- Use the cost-vs-ROI view to flag high-cost, low-ROI campaigns for review rather than assuming bigger budgets automatically improve returns.
- Monitor the click-trend view monthly to catch dips early, since performance appears campaign-driven rather than seasonal.
- Prioritize the top-performing metro markets (Los Angeles, New York, Chicago, Houston) when planning regional campaign budgets.

8. Skills Applied

Data Cleaning, Data Analysis, KPI Design, Dashboard Design, Data Visualization, Marketing Analytics, Storytelling with Data.

9. Conclusion

This project turned a raw, campaign-level marketing dataset into a decision-ready dashboard, surfacing which channels, segments, and locations drive the strongest return. It demonstrates the ability to independently scope, clean, model, and visualize a real-world-style marketing dataset — from KPI definition through to actionable recommendations.